Math 54 — Week 8

Solutions to the problem set

P1. **Idea.** Find eigenvalues and eigenvectors, then use $A^k = PD^kP^{-1}$.

1. The trace is 3 and the determinant is -10, so
$$\det(A - \lambda I) = \lambda^2 - 3\lambda - 10 = (\lambda - 5)(\lambda + 2),$$
and the eigenvalues are 5 and -2.
2. For $\lambda = 5$, $A - 5I$ has first row $(-3, 3)$, giving $x_1 = x_2$: eigenvector $(1, 1)$. For $\lambda = -2$, $A + 2I$ has first row $(4, 3)$, giving $4x_1 + 3x_2 = 0$: eigenvector $(-3, 4)$.
3. Two distinct eigenvalues, so A is diagonalizable with
$$P = \begin{bmatrix} 1 & -3 \\ 1 & 4 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 0 \\ 0 & -2 \end{bmatrix}, \quad P^{-1} = \frac{1}{7}\begin{bmatrix} 4 & 3 \\ -1 & 1 \end{bmatrix}.$$
4. Then
$$A^k = PD^kP^{-1} = \frac{1}{7}\begin{bmatrix} 4 \cdot 5^k + 3(-2)^k & 3 \cdot 5^k - 3(-2)^k \\ 4 \cdot 5^k - 4(-2)^k & 3 \cdot 5^k + 4(-2)^k \end{bmatrix}.$$
For $k = 1$: $\frac{1}{7}(20 - 6,\ 15 + 6;\ 20 + 8,\ 15 - 8) = (2, 3; 4, 1)$, which is A.

P2. **Idea.** Both are triangular or nearly so, so the eigenvalues are visible; the question is the dimension of the eigenspace for the repeated eigenvalue.

1. A is lower triangular with diagonal $2, 2, 3$. The matrix $A - 2I$ has rows $(0, 0, 0)$, $(1, 0, 0)$, $(0, 0, 1)$, so its rank is 2 and $\dim \mathrm{Nul}(A - 2I) = 1$, less than the multiplicity 2. A is not diagonalizable.
2. For B, expand $\det(B - \lambda I)$ down column 2, whose only nonzero entry is $5 - \lambda$:
$$\det(B - \lambda I) = (5 - \lambda)\det\begin{bmatrix} 4 - \lambda & -2 \\ 0 & 5 - \lambda \end{bmatrix} = (5 - \lambda)^2(4 - \lambda).$$
3. $\lambda = 5$: $B - 5I$ has rows $(-1, 0, -2)$, $(2, 0, 4)$, $(0, 0, 0)$, of rank 1, so the eigenspace has dimension 2: $x_1 = -2x_3$ with x_2, x_3 free, basis $(0, 1, 0)$, $(-2, 0, 1)$. $\lambda = 4$: $B - 4I$ gives $x_3 = 0$ and $2x_1 + x_2 = 0$, eigenvector $(-1, 2, 0)$.
4. The dimensions add up to 3, so B is diagonalizable:
$$P = \begin{bmatrix} -1 & 0 & -2 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix},$$
and $BP = PD$.

P3. **Idea.** Count eigenspace dimensions: each simple eigenvalue gives exactly 1, and the repeated one gives $4 - \mathrm{rank}(A - I)$.

The eigenvalues 2 and 3 have multiplicity 1, so their eigenspaces have dimension 1. If $\mathrm{rank}(A - I) = 3$, then $\dim \mathrm{Nul}(A - I) = 1$, and the total is $1 + 1 + 1 = 3 < 4$: not diagonalizable. If $\mathrm{rank}(A - I) = 2$, then $\dim \mathrm{Nul}(A - I) = 2$, the total is 4, and A is diagonalizable.

P4. **Idea.** The columns of the matrix are the images of the $\mathcal{B}$ vectors, in $\mathcal{C}$-coordinates.

1. $T(1) = 1 + t$ and $T(t) = t + t^2$, with $\mathcal{C}$-coordinates $(1, 1, 0)$ and $(0, 1, 1)$. So
$$M = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}.$$
2. $[\,3 - 2t\,]_{\mathcal{B}} = (3, -2)$, and $M(3, -2) = (3, 1, -2)$, so $T(3 - 2t) = 3 + t - 2t^2$.
3. Check directly: $(t + 1)(3 - 2t) = 3t - 2t^2 + 3 - 2t = 3 + t - 2t^2$.

P5. **Idea.** The $\mathcal{B}$-matrix is $P^{-1}AP$; or compute $A\mathbf{b}_1$ and $A\mathbf{b}_2$ and write them in $\mathcal{B}$-coordinates.

$A(1, 1) = (2, 2) = 2(1, 1)$ and $A(1, 2) = (1, 2) = 1 \cdot (1, 2)$. So the $\mathcal{B}$-coordinates of the images are $(2, 0)$ and $(0, 1)$, and the $\mathcal{B}$-matrix is $\operatorname{diag}(2, 1)$. It is diagonal because $(1, 1)$ and $(1, 2)$ are eigenvectors of A, with eigenvalues 2 and 1.

P6. **Idea.** Use one row of $A - \lambda I$ to get the eigenvector, then Thm. 9 with $\lambda = a - bi$.

1. $\det(A - \lambda I) = \lambda^2 - 4\lambda + 5$, so $\lambda = 2 \pm i$.
2. For $\lambda = 2 - i$, the second row of $A - \lambda I$ is $(1,\ 1 + i)$, giving $x_1 + (1 + i)x_2 = 0$. With $x_2 = 1$: $\mathbf{v} = (-1 - i,\ 1)$.
3. $\operatorname{Re}\mathbf{v} = (-1, 1)$, $\operatorname{Im}\mathbf{v} = (-1, 0)$, and $a = 2$, $b = 1$:
$$P = \begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix}, \qquad C = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}, \qquad A = PCP^{-1}.$$
4. The scale factor is $r = |2 - i| = \sqrt{5}$ (also $\sqrt{\det A} = \sqrt{5}$), and the rotation angle has $\cos\varphi = 2/\sqrt{5}$, about $26.6°$.

P7. **Idea.** Take inverses or transposes of $A = PDP^{-1}$; the middle factor stays diagonal.

1. Let $A = PDP^{-1}$. Since $\det A = \det D = d_1 \cdots d_n \neq 0$, every $d_i \neq 0$, so $D^{-1} = \operatorname{diag}(1/d_1, \ldots, 1/d_n)$ exists.
2. The inverse of a product is the product of the inverses in reverse order:
$$A^{-1} = (P^{-1})^{-1}D^{-1}P^{-1} = PD^{-1}P^{-1},$$
so A^{-1} is diagonalizable (same P).
3. Transpose (reverse order again):
$$A^{\mathsf{T}} = (P^{-1})^{\mathsf{T}}D^{\mathsf{T}}P^{\mathsf{T}} = QDQ^{-1} \quad \text{with } Q = (P^{\mathsf{T}})^{-1},$$
because $D^{\mathsf{T}} = D$ and $Q^{-1} = P^{\mathsf{T}}$. So yes, A^{T} is diagonalizable.

P8. **Idea.** Write $\mathbf{x}_0$ in the eigenvector basis; then each piece is multiplied by its eigenvalue at every step.

1. $A(1, 1) = (1, 1)$ and $A(-1, 1) = \frac{1}{2}(-1, 1)$: eigenvalues 1 and $\frac{1}{2}$.
2. $\mathbf{x}_0 = (1, 0) = \frac{1}{2}(1, 1) - \frac{1}{2}(-1, 1)$.
3. So
$$\mathbf{x}_k = \frac{1}{2}(1, 1) - \frac{1}{2}\Big(\frac{1}{2}\Big)^k(-1, 1) = \Big(\frac{1}{2} + \frac{1}{2^{k+1}},\ \frac{1}{2} - \frac{1}{2^{k+1}}\Big).$$
For $k = 1$ this is $(\frac{3}{4}, \frac{1}{4}) = A\mathbf{x}_0$.
4. As $k \to \infty$, $(\frac{1}{2})^k \to 0$, so $\mathbf{x}_k \to (\frac{1}{2}, \frac{1}{2})$, the eigenvector for 1 with entries adding to 1.